25-Hour Keyboard: Optimizing Typing Efficiency with Genetic Algorithms and Chorded Input By Vibbodh Somani

1. Abstract

This study analyzes how typing efficiency can be boosted through the use of evolutionary algorithms and chorded input systems. Standard keyboard layouts like QWERTY and QWERTZ were not created for ergonomics and efficiency, and modern typing demands more optimal alternatives. This study proposes a technique that produces personalized keyboard layouts and abbreviation-based chord systems tailored to individual professions, such as Software Engineering or Authors, and specific languages, primarily focusing on English and German. Using vast data sets and a fitness-based evolutionary technique, the system builds layouts that reduce finger movement and optimize hand utilization. Additionally, a web-based testing tool was built to evaluate the performance of the chord system through user experimentation. The results reveal an average typing speed improvement of 18%, with some users reporting much greater speed improvements with the potential to save up to one an hour a day. Statistical analysis demonstrated the relevance of these improvements. This study demonstrates how keyboards can be redesigned with human centric inputs to provide faster and a more ergonomics typing experience.

2. Introduction

Keyboards have become the primary means of human-computer interaction, whether on computers or mobile devices. Despite their essential function, the predominant keyboard layouts, QWERTY and QWERTZ, were never developed with ergonomics, efficiency, or present usage in mind. They were originally developed for typewriters and primarily for the transcription of Morse code [1].

Modern users allocate roughly 2 ± 1.9 hours daily to keyboard typing and 1.8 ± 2.2 hours to typing on phones. Moreover, a study[3] has demonstrated humans type in letters while thinking in words. This discrepancy impedes communication and exacerbates the fatigue associated with prolonged typing sessions.

This study aims to combine natural cognition with digital input by improving keyboard layouts and incorporating a personalized chorded typing system. The research utilizes computational methods, including genetic algorithms and extensive data analysis, to create customized keyboard systems that cater to users' requirements and improve both efficiency and ergonomics.

3. Related Works

3.1 Existing Keyboard Layouts

Widely established keyboard layouts such as QWERTY and QWERTZ remain in use today not because they are the most efficient, but largely owing to historical convention and general familiarity. In pursuit of better typing efficiency and ergonomics, researchers and niche

communities developed alternative layouts like Dvorak, which can minimize finger movement by over 90%[4], and Colemak, which reduces finger movement by more than 50%[5] when compared to standard layouts such as the QWERTY. These layouts also allow more balanced typing workload distribution between both hands. However, despite their evident ergonomic advantages, these alternatives have failed to achieve broad adoption—likely due to the cognitive and practical expense associated with retraining muscle memory and adapting to a new structure.

3.2 Stenography & Shorthands

To avoid the hassle of typing out long words in full, court reporters and journalists have traditionally employed stenography: a writing technique utilizing shorthand acronyms to denote words or phrases. These abbreviations enable efficient typing in real time. Stenographers frequently utilize a specialized device known as a stenotype machine, which has piano-like keys. Analogous to the simultaneous pressing of many piano keys yielding a distinct chord, the simultaneous pressing of several stenotype keys results in a particular shorthand.

This study investigates the implementation of stenographic principles on a standard keyboard. The use of a personalized shorthand system, termed a chord system, enables users to improve their typing speed by inputting shorthands that instantly convert into complete words or sentences. This method incorporates the speed benefits of stenography into conventional typing settings without requiring additional equipment.

4. Theoretical Background

4.1 Basics of Keyboard Layout Optimization

In this study, an optimized keyboard layout is defined as one that minimizes finger movement, balances typing effort across both hands, and avoids repetitive use of the same finger for consecutive characters. Depending on the user, layouts may be symmetrical or biased toward the dominant hand or stronger fingers. To quantify these factors, we used an open-source dataset [6] that provides two key metrics:

1. Effort required to press each key
2. Distance traveled from the home position to reach each key

Diagram 1: Typing Effort by Finger

Diagram 1 illustrates the relative effort needed to press each key. Each color represents a finger's typing zone:

- Left hand: pinky (orange), ring (yellow), middle (light green), index (dark blue)
- Right hand: index (light blue), middle (purple), ring (pink), pinky (red)
- Thumbs: dark green

The numbers in each key indicate effort values. Home row positions generally have the lowest effort (value 0), except for the pinky, which starts at effort level 1. The diagram shows that despite equal distances, pressing keys with weaker fingers (like the pinky) requires more effort than with stronger fingers (like the ring). This reflects differences in finger dexterity.

Diagram 2: Physical Distance from Start Position

Diagram 2 shows the same finger zones, but instead of effort, the numbers indicate the physical distance each key is from its home position. This metric represents the actual movement required, independent of finger strength.

4.2 Genetic Algorithms

Genetic algorithms (GAs)[7] are a class of evolutionary optimization techniques inspired by the principles of natural selection and genetics. They are particularly helpful at tackling complex, multi-variable problems where exact solutions are computationally challenging. While GAs may not guarantee perfect answers, they are well-suited for identifying high-quality, near-optimal results fast.

A genetic algorithm begins with the initialization of a population of random candidate solutions. Each individual is then evaluated using a fitness function that measures its quality with respect to the problem. The best-performing candidates are selected to act as parents, and their traits are combined through crossover to produce new offspring. To maintain genetic diversity, mutations, small random changes, are introduced into some offspring. The resulting individuals replace the old population, forming a new generation. This cycle of evaluation, selection, crossover, mutation, and replacement is repeated until termination, which occurs when the algorithm converges or meets a predefined stopping condition.

4.3 Principles of Chorded Typing Systems

The chorded typing system developed in this study aims to improve typing efficiency by minimizing the number of keystrokes typed. It is molded by the ideas of stenography and shorthands and builds upon two primary strategies:

A. Simplification through Conventions and Shortcuts

Many terms are replaced with shorter, standardized abbreviations or mnemonic shortcuts. This is prevalent in professional domains where speedy data entry is vital. For example:

- "information" → info
- "documentation" → doc
- "diagnosis" → dx
- "prescription" → rx

These shorthands follow broadly accepted norms or are user-defined abbreviations that favor minimal input for maximum efficiency. Users can also create their own unique chords for frequently used words or phrases

B. Letter Elimination (Inspired by Journalistic Shorthand)

Another technique involves progressively deleting vowels and unnecessary letters to generate compressed word versions, while keeping recognizability. This mirrors approaches used in shorthand systems adopted by journalists and stenographers. For example:

- "through" → thrgh
- "development" → dvlpmnt
- "project" → prjct
- "because" → bcs

By deleting vowels and reducing phonetics, these chord forms can be entered faster while still being easily remembered by the user.

5. Methodology

5.1 Design of Genetic Algorithm

In the context of this study, the GA is used to optimize keyboard layouts for increased typing ergonomics and speed. The method begins with the random generation of an initial population of keyboard layouts. The implementation in this study consists of a base population with 100 randomly generated layouts. A fitness function is then applied to evaluate each layout. This function considers various ergonomic and performance parameters, including:

1. Finger travel distance: total movement required for common input patterns.

2. Hand and finger workload distribution: balancing of typing effort across both hands and all fingers.
3. Repetitive strain penalties: penalization of frequent same-finger or same-hand usage patterns.

After evaluation of the top-performing layouts the algorithm selects 10 of the highest-scoring layouts to act as parents for the next generation. These parent layouts undergo a crossover operation, where each child layout is created by taking approximately half of its key assignments from each parent. The split is randomized to maintain diversity.

To imitate genetic mutation and introduce diversity, a limited number of random key swaps are done to each child layout. This helps explore new regions of the solution space and avoid premature convergence to local optima.

Once 100 new layouts have been formed via crossover and mutation, they comprise the next generation, and the evaluation-selection-breeding process is repeated. This iterative evolution continues until the algorithm sees no substantial improvement in fitness across multiple successive generations, signifying convergence. This adaptive, feedback-driven technique enables the GA to progressively uncover layouts that reduce effort and increase ergonomic efficiency.

5.2 Design of Chorded Typing System

The generation of chords for the chorded typing system is largely automated. The first step in this procedure is to identify which words should receive chords—specifically, those that are both commonly used and demand the greatest typing effort due to their length or intricacy. This is measured through the fitness function of the genetic algorithm for the current keyboard layout. These phrases are prioritized for chording as they provide the greatest benefit to the user.

To begin, the text data is tokenized, meaning it is broken into discrete units (tokens) by splitting the text based on whitespace, punctuation marks, or special characters. This procedure enables the system to compute word frequencies across the dataset. However, raw frequency is not sufficient. Additional filtering is used to exclude tokens that are not relevant to the target use case or profession. For instance, URLs, serial numbers, and system-generated identifiers are omitted from the final dataset, as they do not commonly appear in everyday writing.

Once the dataset has been filtered, the system ranks phrases based on both frequency and typing difficulty. The most burdensome words receive the shortest and most efficient chords. These chords are combined into a user profile—a personalized lexicon that matches each abbreviation to its matching word. This profile forms the backbone of the chorded typing experience.

To implement this approach at a practical level, a system wide application was designed that operates across all software environments (e.g., word processors, web browsers). This application listens to keyboard inputs and extends chords into whole words in real time, regardless of the program being used. The system operates as follows:

1. Initialization: The application loads the user's chord profile and starts listening to keyboard input globally.
2. Input Collection: Characters are gathered into a buffer until a delimiter (e.g., spacebar) is pressed.
3. Abbreviation Lookup: The system checks if the buffer matches any defined chord in the profile.
4. Expansion: If a match is found, the abbreviated input is replaced with the full word.
5. Reset: The buffer is cleared to begin collecting the next word.

This implementation enables for easy integration of the chord system across many platforms and text editors. By automating both the generation and deployment of personalized chord dictionaries, the technology greatly decreases typing effort without forcing users to switch applications or employ proprietary hardware.

5.3 Data Sources and Preprocessing

This study focuses on three profiles: English, German, and Programming(profession specific). For each, a domain-specific dataset was collected to train the chorded typing system and optimize the keyboard layout using a genetic algorithm.

1. English Profile: A corpus of 30,000 English books was scraped from Project Gutenberg, spanning various genres and publication eras to ensure linguistic diversity.
2. German Profile: A set of 1,400 German books was compiled from Project Gutenberg and other online sources, covering a range of writing styles and historical periods.
3. Programmer Profile: 26 million lines of Python code were collected from public code repositories and forums to reflect realistic coding patterns.

5.4 Evaluation Metrics & Tests

To evaluate the effectiveness of the optimized keyboard layouts and the chorded input system, two separate methodologies were utilized: simulation and user testing.

Evaluation of Keyboard Layouts (Simulation-Based)

The optimized keyboard layouts were evaluated by simulation instead of real user testing. This decision was based on the acknowledgment that learning an entirely new keyboard layout is cognitively taxing and time-consuming. Users are profoundly used to conventional layouts such as QWERTY or QWERTZ, owing to years of muscle memory, rendering a temporary experimental shift impractical.

To tackle this issue, the study employs an external keyboard analysis tool[8] that simulates typing behavior to assess typing effort. The tool measures characteristics such as finger travel distance, hand alternation frequency, and finger load balance. This enabled the study to

calculate an optimal efficiency score for each layout under ideal typing conditions, providing a theoretical maximum performance limit.

Evaluation of the Chorded Typing System (User-Testing)

In contrast, the chorded input system was tested by direct user testing via a custom-built online platform. The platform took participants through the following test flow:

1. Standard Typing Assessment - Typing a sample passage without any abbreviations or chords
2. Chorded Typing Assessment – Inputting identical text while utilizing established chord shortcuts
3. Results Display — A visual comparison of typing times from both tests
4. Optional Feedback – A spot for participants to review their experience or give comments

The text utilized in both examinations was domain-specific, targeted to professions such as medicine, law, or software development. Typing speed was recorded automatically, and efficiency improvements were estimated as a percentage difference between the two tests.

Although the test was purposely brief, it served as a baseline measure of how people performed when initially introduced to the chord system. It is crucial to highlight that participants had access to a help sheet for the chord definitions, and they encountered the system for the first time. Over time, as users internalize the chord mappings, their performance is expected to increase significantly.

The short and accessible nature of the test allowed the study to acquire data from a broader and more diverse participant pool, while still offering valuable insights into the system's potential efficiency gains.

6. Results

6.1 Comparison of Keyboard Layouts

To evaluate the performance of different keyboard layouts, we conducted a simulation-based analysis using a weighted scoring system. The evaluation considered three key ergonomic and efficiency-related factors:

1. Finger travel distance (33%)
2. Finger usage frequency (33%)
3. Hand and finger alternation frequency (34%)

Each layout received a composite score, with higher scores indicating better overall performance in reducing physical strain and improving balance across the hands and fingers.

English Layout Comparison

The English keyboard analysis included:

1. Our optimized layout
2. Dvorak
3. Colemak
4. Standard QWERTY

The optimized layout achieved the best performance with the highest efficiency score and the lowest standard deviation (1.34), showing strong ergonomic balance. Dvorak followed with a deviation of 1.69, then Colemak with 3.49. QWERTY again performed the worst with a standard deviation of 3.79.

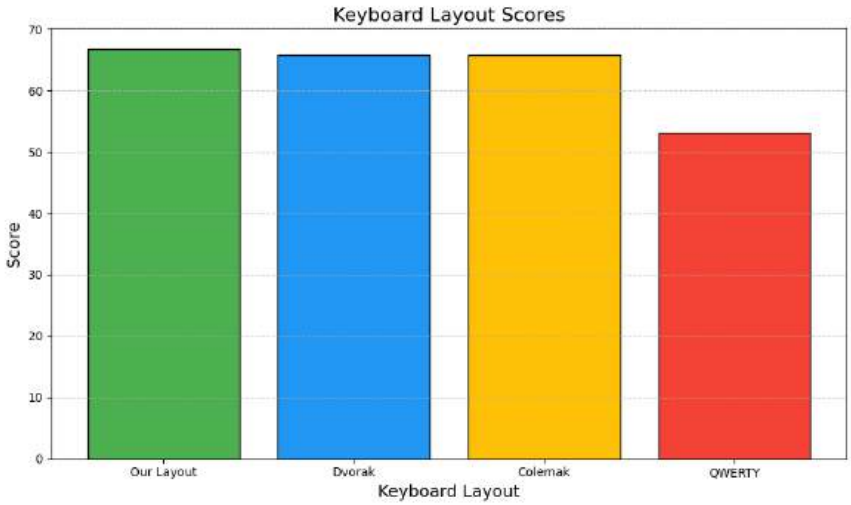

Diagram 3: Keyboard Scores through English Typing Simulation

To provide a more visual perspective, the generated heatmaps show the distribution of key usage across the QWERTY and optimized English layouts. In the QWERTY heatmap, high-usage areas (hot zones) appear scattered across the top, middle, and bottom rows of the keyboard. This spread indicates significant finger movement across all rows, contributing to fatigue and inefficiency. In contrast, the optimized layout shows that the majority of key presses are concentrated on the home row, with minimal movement to other rows. This visual pattern confirms that the optimized layout achieves its goal of reducing finger travel and centralizing input activity.

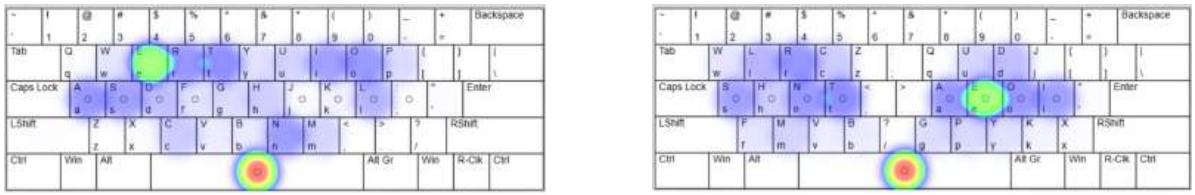

Diagram 4:Heatmap of Key-Usage in QWERTY vs Optimized English Layout

German Layout Comparison

For the German language, four layouts were compared:

1. Our genetically optimized layout
2. Dvorak German
3. AdNW (Aus der Neo-Welt)
4. Standard QWERTZ

AdNW achieved the highest overall score and the lowest standard deviation in finger usage (2.90), indicating an especially balanced load across the fingers. Our optimized layout also scored highly, though with a slightly higher standard deviation (5.25), suggesting minor imbalances in finger usage. Dvorak German showed moderate performance with a standard deviation of 3.89. QWERTZ performed the worst, both in overall score and balance (6.02), highlighting its inefficiency from an ergonomic standpoint.

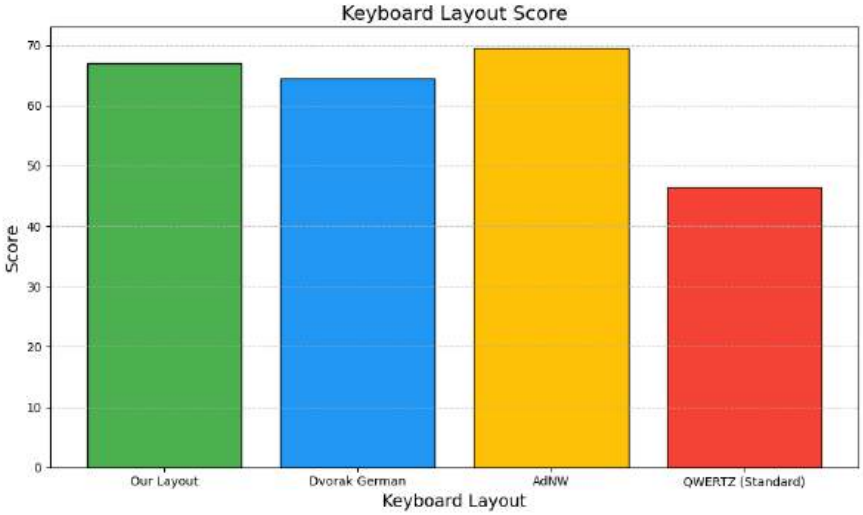

Diagram 5:Keyboard Scores through German Typing Simulation

Programming Layout Comparison (Python Dataset)

To assess layout performance in a real-world technical context, we evaluated common programming layouts using a dataset of Python code. The following layouts were tested:

1. Standard QWERTY
2. Colemak Mod-DH
3. Workman
4. Programmer Dvorak

In this case, lower scores indicate better efficiency. The optimized layout outperformed all other layouts in minimizing finger travel and improving balance for Python-specific syntax. While Programmer Dvorak and Colemak Mod-DH outperformed QWERTY, they still lagged behind the optimized layout.

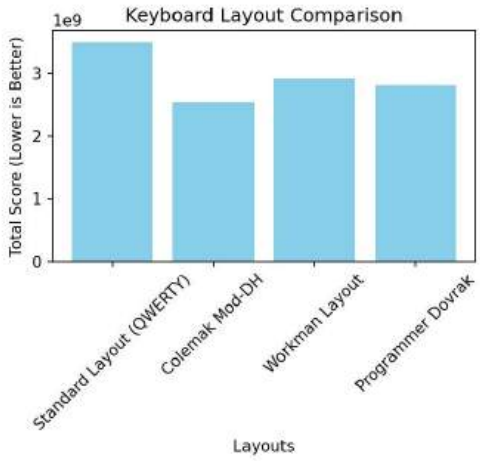

Diagram 6:Keyboard Scores through Python(Programming) Typing Simulation

6.2 Efficiency Gains from Chorded Typing System

The effectiveness of the chorded input system was evaluated through user testing on the custom-built web platform. Participants completed two typing tasks, once using regular typing and once using the chorded system, and the platform automatically logged typing times. The first graph below depicts the raw typing times for each participant in both conditions—standard typing and chorded typing. The second graph indicates the % change in typing speed, with positive values indicating improvements and negative values representing poorer performance with chords.

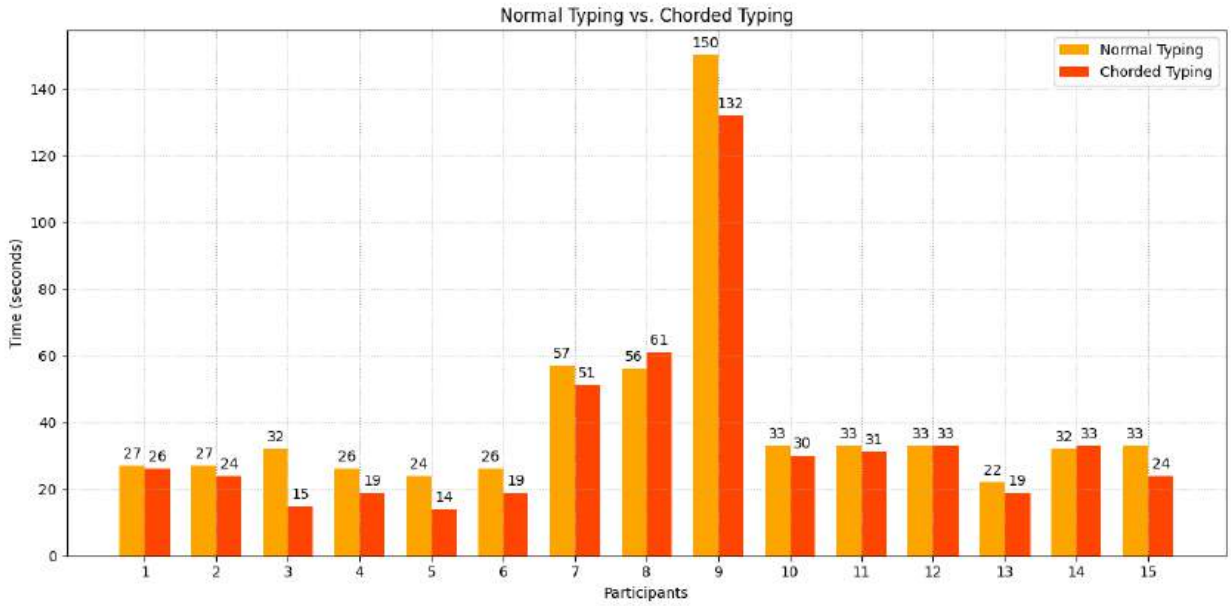

Diagram 6:Comparison typing time – Regular typing vs. chord typing

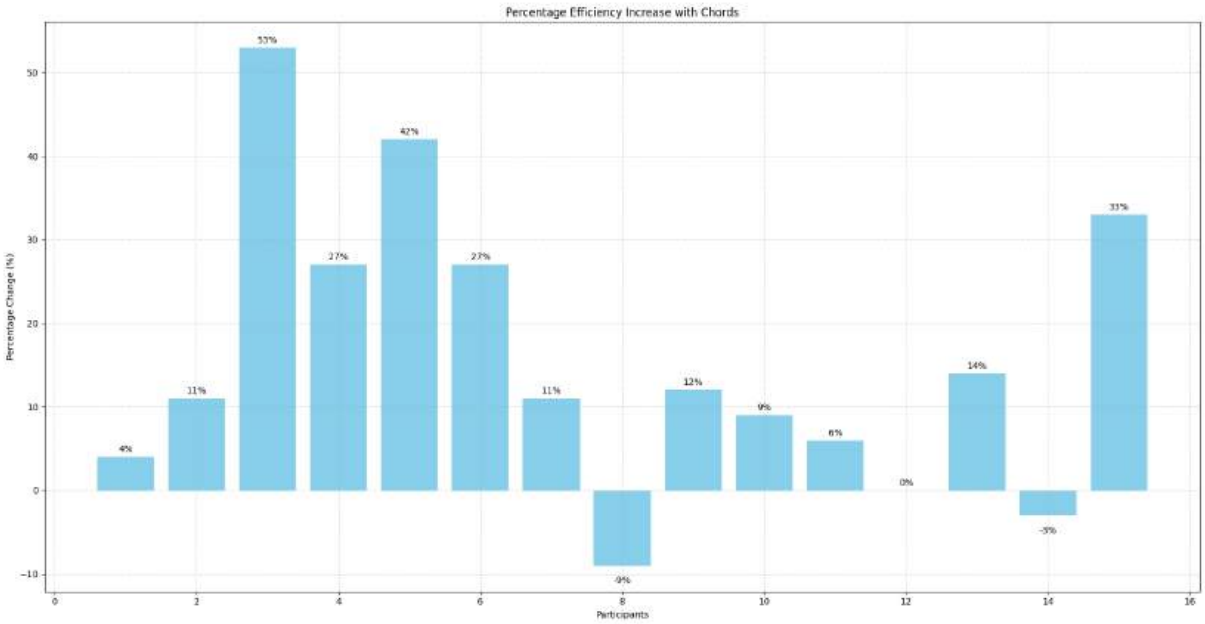

Diagram 7:Percentage increase (or decrease) in typing efficiency per participant after using chorded input.

Overall, our data demonstrate a clear positive link between the use of chorded typing and better typing efficiency. Most individuals demonstrated considerable gains in speed. A few individuals showed reduced efficiency while employing chords, which they attributed to unfamiliarity with the system rather than problems in the method itself. These users remarked that with practice, the chords could become substantially more helpful.

Across all participants, the average efficiency improvement was 18%, with a median improvement of 12%. Two people observed decreasing efficiency, whereas thirteen participants demonstrated increased efficiency.

With additional time and testing resources, repeated trials across different text samples to account for learning effects and assess retention of chord knowledge. Nevertheless, the results reveal strong potential for substantial time savings with regular usage.

7. Discussion

7.1 Interpretation & Practical Implications of Results

The findings of this study indicate that substantial enhancements in typing speed and ergonomics can be attained without modifying the physical keyboard. Combining optimal layouts with a chorded input system can yield a typing efficiency enhancement of more than 20% for users.

These changes have significant consequences for professions who depend extensively on typing in their everyday tasks—especially physicians, attorneys, software developers, and

authors. The system's adaptability and capacity for personalization render it particularly advantageous for persons in high-pressure environments where time is critical.

The observed efficiency improvements indicate a prospective time reduction of roughly one hour daily by minimizing redundant keystrokes and superfluous finger movements. This enhances productivity while decreasing physical strain, rendering prolonged typing sessions more sustainable and comfortable over time.

7.2 Limitations of the Study

This study faced multiple limitations that constrained the breadth of its findings. The optimized keyboard layouts were assessed solely through computational simulations, without real-world user testing. As a result, the study is unable to analyze the learning curve or cognitive effort required for users to adopt and become effective with different layouts. Understanding how individuals adjust to different layouts over time would provide useful insights into their practical usability and effectiveness.

Secondly, while the chorded input system exhibited promising short-term results, its long-term impact remains unknown. The current evaluation concentrated on immediate efficiency advantages, without examining how user familiarity with the system builds over extended use. It is possible that, as users become more acclimated to the chords, their typing speed would increase further—especially as they rely less on reference materials to recall individual chords.

Another important issue was the difficulty in getting domain-specific datasets, such as real texts authored by professionals like doctors or attorneys. The lack of such data prohibited a further exploration into individualized improvements geared to specific occupations. As a result, the systems built in this study were trained predominantly on general-purpose datasets. This study predicts that more specialized datasets could enable even greater increases in efficiency through more targeted tailoring.

7.3 Future Work

Future research should explore the learning curves associated with both the optimized keyboard layouts and the chorded typing system in greater detail. Understanding how quickly users can adapt to these systems—and how long the benefits take to materialize—is essential for evaluating their long-term usability.

Additionally, future studies would benefit from conducting more extensive experimental evaluations using real-world data and long-term usage scenarios. This would provide deeper insight into the practical challenges and cognitive load involved in adopting such systems.

It is also recommended that researchers further refine the use of genetic algorithms by experimenting with a broader range of parameters and optimization strategies. Identifying optimal configurations early in the development process could significantly enhance the effectiveness and efficiency of the final keyboard layouts.

8. Conclusion

This study introduced the 25-Hour Keyboard, a system designed to enhance typing efficiency through the use of genetic algorithms and a chorded input system. By addressing the inefficiencies of legacy keyboard layouts such as QWERTY and QWERTZ, and drawing inspiration from stenography and shorthand techniques, the project successfully demonstrated how modern computational methods can be used to design personalized, high-performance keyboard systems.

Through simulation and user testing, the project revealed that users can experience an average improvement of 18% in typing speed using the chorded input system, with some reporting substantially higher gains. Simulations of optimized layouts showed reduced finger travel, improved ergonomic balance, and better distribution of workload across both hands compared to standard and alternative layouts.

The implications of this work are especially significant for professionals who spend large portions of their day typing—such as software developers, medical professionals, and legal practitioners. By reducing keystrokes and physical strain, this system not only improves speed but also offers long-term ergonomic benefits.

While limitations remain, particularly regarding long-term user adoption and access to domain-specific data, this project provides a strong foundation for future research. It underscores the value of human-centric design in digital tools and showcases how evolutionary computation can meaningfully enhance everyday human-computer interaction.

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